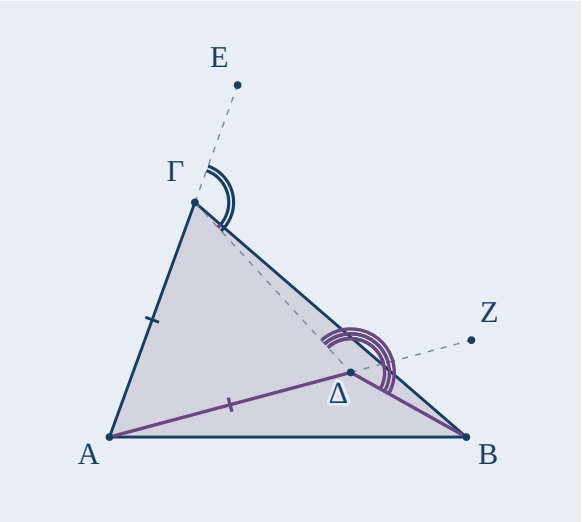
The case left to the reader

Δ inside the triangle



Control: 55° turn from Γ toward the base. AΓ = AΔ is retained. The second distance equality is a supposition in the contradiction argument, not a fact of a distinct same-side sketch.

What this sketch actually satisfies

Condition	Measured sketch
AΓ = AΔ	140.00 = 140.00
BΓ = BΔ	201.11 ≠ 74.22
Same side of AB	Yes
Distinct Γ and Δ	Yes
∠ΔΓB part of ∠AΓΔ?	No · use exterior angles
∠ΔΓB / ∠ΓΔB	6.65° / 161.72°

The printed part relation no longer fits

At Γ, ray ΓB lies outside ∠AΓΔ. So ∠ΔΓB is not a part of ∠AΓΔ. Their numerical sizes can still have the desired order; that does not repair this reason for the order. We need a different containing angle.

$$\angle A\Gamma\Delta = \angle A\Gamma B - \angle \Delta\Gamma B$$

Use the angles below the base

Extend AΓ beyond Γ to E, and AΔ beyond Δ to Z (postulate 2). In isosceles AΓΔ, I.5 makes the two exterior angles EΓΔ and ZΔΓ equal. E and Z are labels for this supplement; they do not occur in the printed I.7 passage.

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∠EΓΔ = ∠ZΔΓ (I.5, exterior angles)
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Carry the order through equal angles

At Γ, ∠ΔΓB is now a proper part of ∠EΓΔ. At Δ, ∠ZΔΓ is a proper part of ∠ΓΔB. The equal exterior angles sit between the smaller and larger angles. This uses whole and part, substitution of equal magnitudes, and transitivity of order.

```
∠ΔΓB < ∠EΓΔ = ∠ZΔΓ < ∠ΓΔB
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The second equality would contradict this order

If BΓ also equalled BΔ, I.5 in triangle BΓΔ would make ∠ΔΓB equal to ∠ΓΔB. The previous comparison made them unequal. That is the contradiction. The sketch retains the first equality and displays the actual unequal second pair; it cannot draw both incompatible assumptions as facts.

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If BΓ = BΔ: ∠ΔΓB = ∠ΓΔB. But ∠ΔΓB < ∠ΓΔB.
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Read the clause that makes the repair possible

The base of the isosceles triangle AΓΔ is ΓΔ. “Below the base” means the angles made by extending the equal sides past Γ and Δ; it does not mean lower on the screen.

Τῶν ἰσοσκελῶν τριγώνων αἱ πρὸς τῇ βάσει γωνίαι ἴσαι ἀλλήλαις εἰσίν, καὶ προσεκβληθεισῶν τῶν ἴσων εὐθειῶν αἱ ὑπὸ τὴν βάσιν γωνίαι ἴσαι ἀλλήλαις ἔσονται.

In triangulis aequicruriis anguli ad basim positi inter se aequales sunt, et productis rectis aequalibus anguli sub basi positi inter se aequales erunt.

In isosceles triangles the angles at the base are equal to one another; when the equal sides are extended, the angles below the base will also be equal to one another.

ὑπὸ τὴν βάσιν / sub basi: “below the base.” In the angle name ἡ ὑπὸ ΑΓΔ, the letters instead identify the two bounding rays, with Γ as vertex.

I.5 statement: Heiberg (1883), Greek p. 20; Latin p. 21. The Latin is Heiberg’s nineteenth-century translation. English: Stoicheia.

Modern exploration, not an additional Greek or Latin passage from I.7. Heath's commentary (1925), pp. 259–261, discusses the omitted interior case and attributes it to Proclus. This apparatus uses fixed AB = 200, AΓ = 140, and ∠BAΓ = 70° to cross the relevant placements continuously. It does not enumerate all placements or supply a formal axiomatization of order. Measurements check the real-plane drawing; they are not its proof.

Heath commentary, transcribed by D. R. Wilkins · David E. Joyce, I.7 commentary · Heiberg source volume

Greek and Latin source: public domain. English explanations and calculated diagrams: Stoicheia (2026), MIT.